Assignment 5: Data Visualization

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy `NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv` version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the `NEON_NIWO_Litter_mass_trap_Processed.csv` version, again from the Processed_KEY folder).

2. Make sure R is reading dates as date format; if not change the format to date.

```
#1 Workspace set up, loading packages, checking WD and importing files
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.0     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```r
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```r
# Checking WD
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```r
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```r
# Importing datasets

Peter.Paul <- read.csv(here(
  "./Data/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv"), stringsAsFactors =

Niwo <- read.csv(here(
  "./Data/Processed_KEY/NEON_NIWO_Litter_mass_trap_Processed.csv"),
  stringsAsFactors = TRUE)

#2 Changing formats to date
Peter.Paul$sampledate <- ymd(Peter.Paul$sampledate)
Niwo$collectDate <- ymd(Niwo$collectDate)
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```r
#3 Making a custom theme

eggyolk_theme <- theme_classic() +
  theme(
```

```
    axis.title = element_text(color = "orange"),
    legend.position = "right",
    plot.title.position = "plot",
    plot.background = element_rect(color = "gray")
  )

theme_set(eggyolk_theme)
```

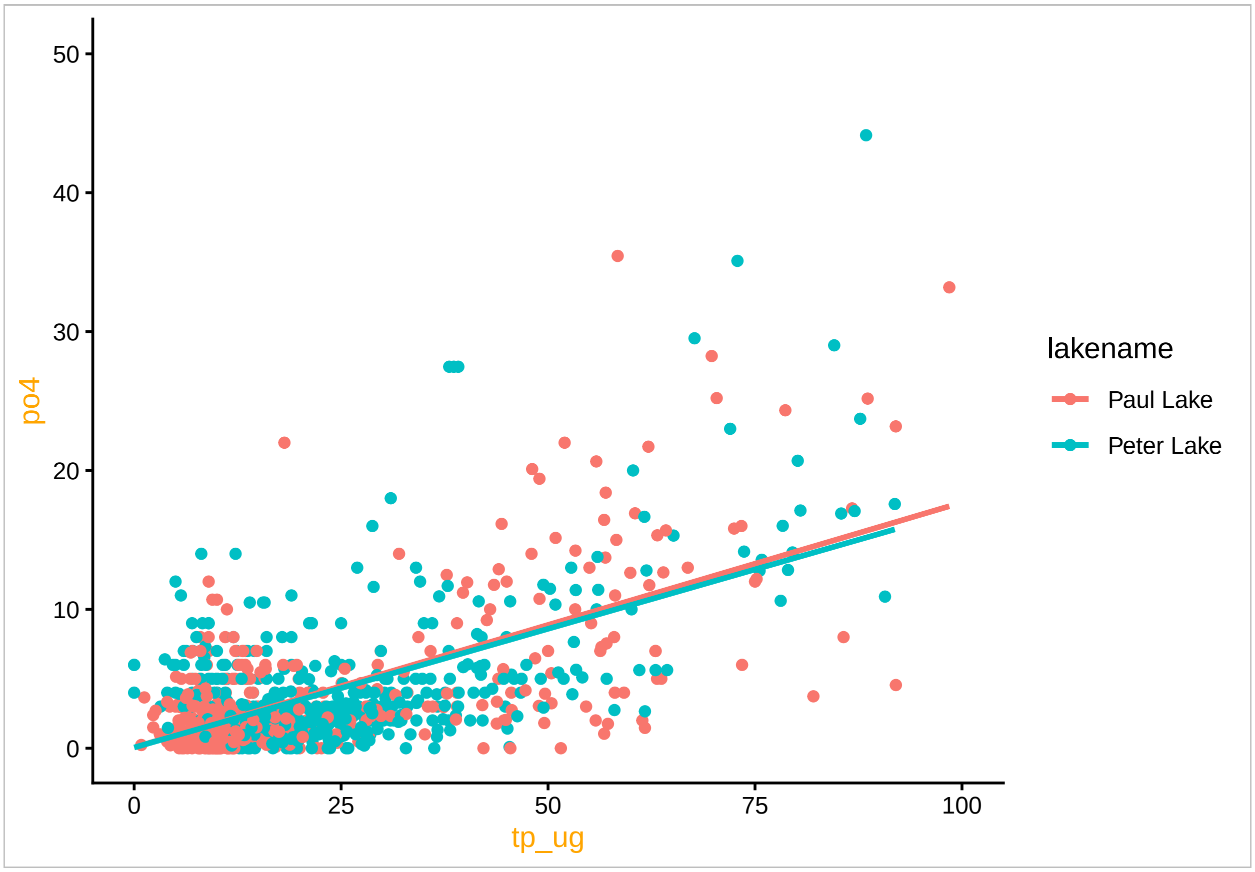
Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (`tp_ug`) by phosphate (`po4`), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

```
#4 Plotting total phosphorus by phosphate for Peter and Paul Lakes

Peter.Paul %>%
  ggplot(
    aes(
      x = tp_ug,
      y = po4,
      color = lakename)
    ) +
  geom_point() +
  geom_smooth(
    method = "lm",
    se = FALSE
) +
  xlim(0,100)+
  ylim(0,50)
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 21964 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 21964 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

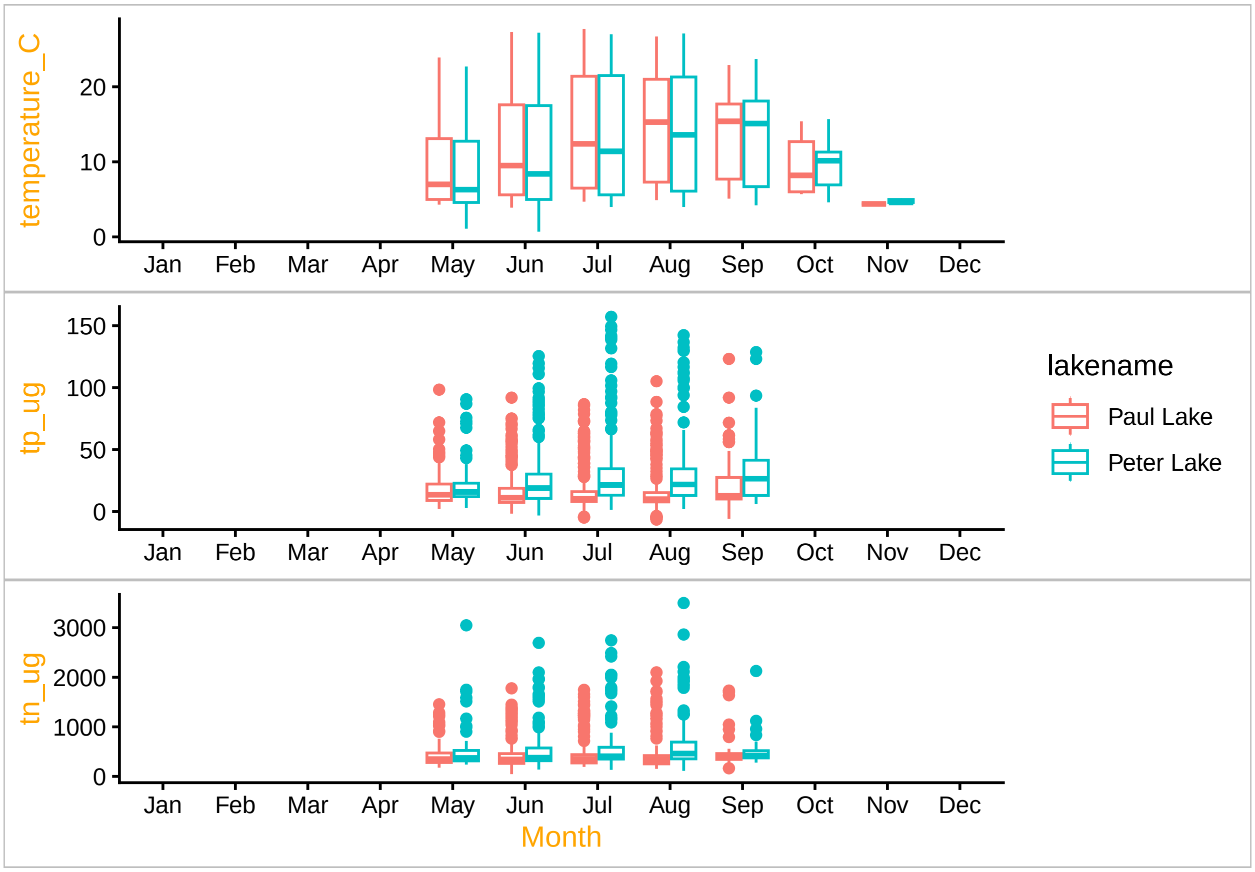
Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

```
#5
#commenting out to allow for knitting
#factor(Peter.Paul$month,
#        levels = 1:12,
#       labels = month.abb)

tempplot <- ggplot(Peter.Paul,
    aes(
      x = factor(
        month,
        levels = 1:12,
        labels = month.abb),
      y = temperature_C,
      color = lakename)) +
    geom_boxplot()+
  scale_x_discrete(
```

```
    name = 'Month',
    drop = FALSE
  ) +
  theme(
    axis.title.x = element_blank(),
    legend.position = "none",
    )


TPplot <- ggplot(Peter.Paul,
    aes(
      x = factor(
        month,
        levels = 1:12,
        labels = month.abb),
      y = tp_ug,
      color = lakename)) +
    geom_boxplot()+
  scale_x_discrete(
    name = 'Month',
    drop = FALSE
  ) +
   theme(
    axis.title.x = element_blank(),
    ylab("TP")
  )


TNplot <- ggplot(Peter.Paul,
    aes(
      x= factor(
        month,
        levels = 1:12,
        labels = month.abb),
      y = tn_ug,
      color = lakename)) +
    geom_boxplot() +
  scale_x_discrete(
    name = 'Month',
    drop = FALSE
  ) +
   theme(
     legend.position = "none",
     ylab("TN")
   )


plot_grid(tempplot, TPplot, TNplot, nrow = 3, align = "v")
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning: Removed 20729 rows containing non-finite outside the scale range
```

```
## ('stat_boxplot()').

## Warning: Removed 21583 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



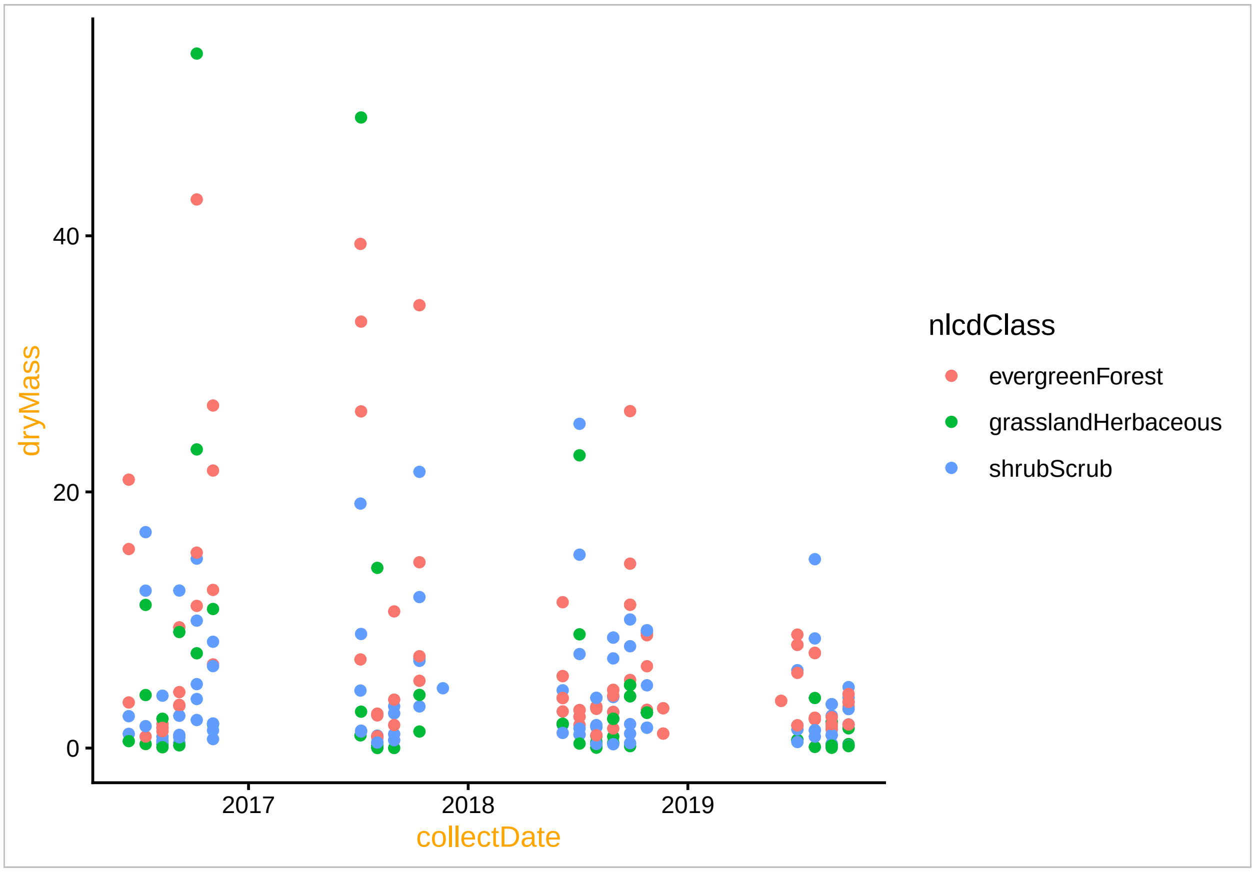
Question: What do you observe about the variables of interest over seasons and between lakes?

> Answer: For temperature, we see a fairly steady increase in both lakes from May through September and then a drop in October. This trend makes sense given seasonal changes in temperature. Phosphorous seems to remain fairly steady throughout the year for both lakes, though Peter Lake did have a slight uptick as the months went on and is generally displaying higher levels than Paul Lake. The same is true for nitrogen, though perhaps to a lesser extent.

6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the "Needles" functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

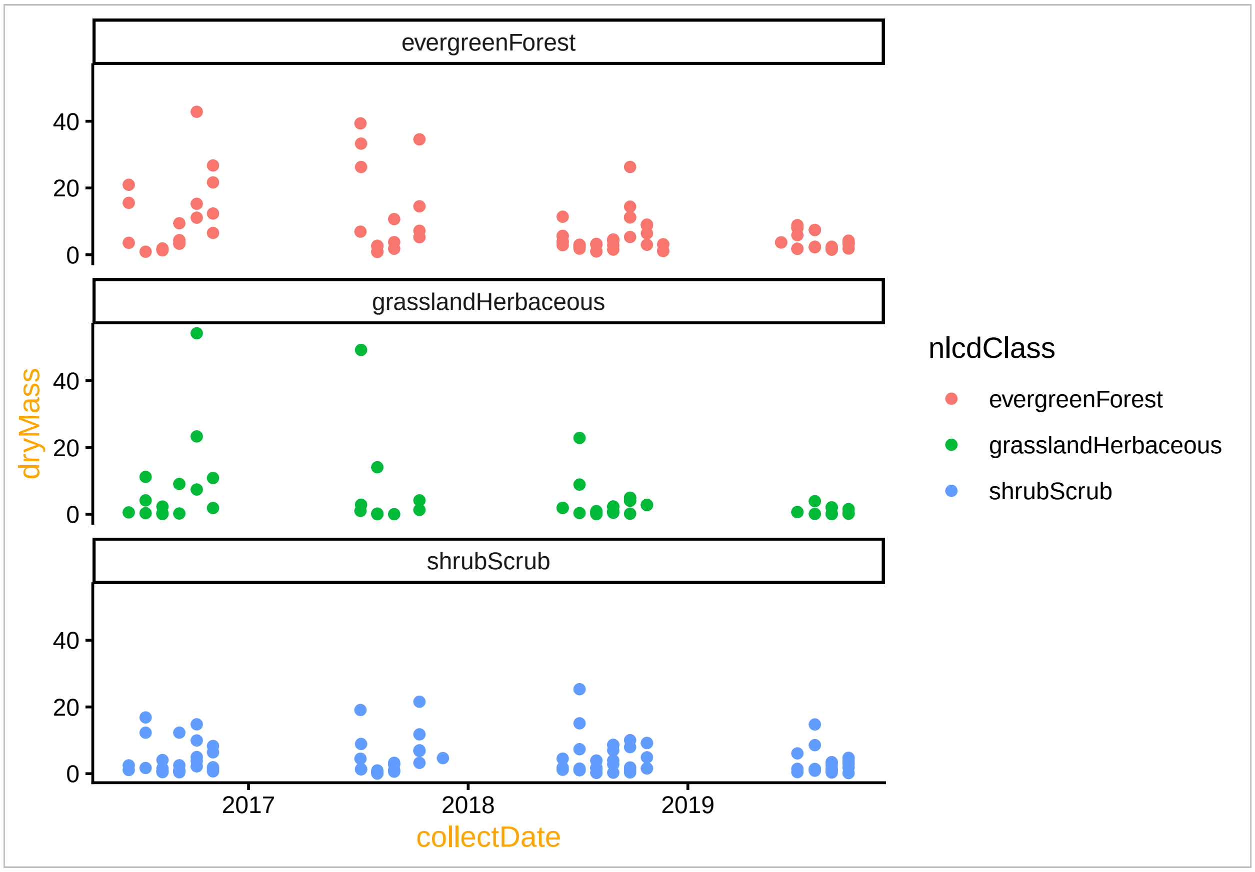
```
#6
Niwo.litter <- Niwo %>%
  filter(
    functionalGroup == "Needles") %>%
  ggplot(
```

```
    aes(
      x = collectDate,
      y = dryMass,
      color = nlcdClass
    )
  ) +
  geom_point()

print(Niwo.litter)
```




```
#7
Niwo.litter.faceted <- Niwo %>%
  filter(
    functionalGroup == "Needles") %>%
  ggplot(
    aes(
      x = collectDate,
      y = dryMass,
      color = nlcdClass
    )
  ) +
  geom_point()+
  facet_wrap(
    facets = vars(
      nlcdClass),
```

```
      nrow = 3)

print(Niwo.litter.faceted)
```



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

> Answer: I find 7 to be much more effective as I can more easily see any potential trends within a specific class. In 6, the data points are all grouped so closely together that it's hard to see any meaningful patterns that might emerge depending on the color/class. When they are separate and distinct, I can see more clearly how/if things change over the years.